

CHRIS BHAILA

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OBJECTIVE

AI/ML engineer with hands-on experience across the full ML pipeline — data collection, cleaning, preprocessing, model building, and evaluation. Seeking an internship building and iterating on ML models that ship into real applications, grounded in careful evaluation and clear documentation.

EDUCATION

Bachelor's in Computer Application 2021 - 2026

Thames International College, Kathmandu

Relevant Coursework: Web Technology, MySQL, Scripting Language, Computer Graphics, Data Structures and Algorithms, Probability and Statistics, Digital Logic, Numerical Methods, DotNet Technology, OOP in Java, Financial Accounting, Software Project Management, Artificial Intelligence

+2 2019 - 2021

Management with Computer Science

Little Angels' College of Management, Hattiban, Lalitpur, GPA: 3.38

Relevant Coursework: Economics, Calculus, C Programming

SKILLS

AI/ML	PyTorch, scikit-learn, Sentence-BERT/Transformers, NumPy, Pandas, Keras, Model Evaluation & Iteration
Programming	Python, JavaScript, PHP, Java, C, C#
Data	Web Scraping (BeautifulSoup), Data Cleaning, MySQL, PostgreSQL, Matplotlib, Plotly
Web/Backend	Django, Laravel, REST APIs
Tools	Git, Jupyter, MATLAB
Languages	English, Nepali, Newari

INTERNSHIP

Full-Stack Developer Intern 2026

Nahusha Investment Pvt. Ltd.

Kathmandu, Nepal

- Developed FloraNepal (a.floranepal.com), a nursery management web application, as the primary project deliverable using Laravel, Blade, Tailwind CSS, and Alpine.js
- Implemented token-based authentication with Laravel Sanctum and Google SSO via Laravel Socialite, enabling secure multi-provider user login
- Engineered an SPA-like navigation pattern using Alpine.js and custom **X-Dashboard-Navigate** headers, reducing full-page reloads across the dashboard

PROJECTS

Traffic Hotspot Detection System 2026

 [GitHub](#)

- Implemented DBSCAN from scratch using Haversine distance to detect 43 accident hotspots across Leeds, UK (1,498 records), with epsilon selected automatically via the k-distance elbow method rather than hardcoded
- Built an ID3 decision tree from scratch (no scikit-learn) for accident severity prediction, achieving 63.5% accuracy; evaluated results and identified class imbalance (14 Fatal records vs 1,062 Slight) as the primary failure mode
- Developed dual-language scrapers for English (Kathmandu Post) and Nepali (OnlineKhabar) accident articles using Devanagari keyword dictionaries for severity, time, and weather extraction

- Deployed full-stack Django application on Render with PostgreSQL, serving an interactive Leaflet.js hotspot map and Chart.js analytics dashboard

Image Classification with Convolutional Neural Networks

2026

- Implemented and compared CNN architectures from scratch (LeNet, ResNet, AlexNet) in PyTorch on the MNIST dataset, evaluating accuracy, precision, recall, and F1-score
- Applied transfer learning with pretrained VGG16 and EfficientNetB0 (Keras/TensorFlow) for multi-class image classification, including partial fine-tuning of top backbone layers to improve accuracy on a domain-specific dataset

Product Web Scraper & Data Pipeline (Bass Central / Yamaha)

2026

- Built a web scraper using BeautifulSoup to extract product data and images from e-commerce sources, exporting results to CSV
- Cleaned and transformed scraped data with Pandas, then visualized trends using Matplotlib and Plotly
- Designed a MySQL storage pipeline to persist cleaned product data, with Python handling the transfer from DataFrame to database

AI-powered Recommendation System for E-commerce Store

2025

 [GitHub](#)

- Built a semantic retrieval engine using Sentence-BERT embeddings and cosine similarity to surface contextually relevant products
- Built full-stack e-commerce platform (PHP/Python, MySQL, HTML/CSS) with user authentication, inventory management, and item filtering
- Integrated a user rating system to capture feedback for future recommendation refinement

IntegrityCheck – Academic Plagiarism & Similarity Detection Platform

2026

 [GitHub](#)

- Built a dual-signal similarity detection engine from scratch, combining n-gram shingling with Jaccard similarity (lexical) and Sentence-BERT embeddings with custom cosine similarity (semantic) into a weighted combined score, catching both verbatim copying and reworded paraphrasing
- Extended detection beyond peer submissions to the open web via the Google Custom Search API and a BeautifulSoup scraping pipeline, routing scraped page text through the same similarity pipeline as student-vs-student comparisons
- Designed a role-based Laravel platform (student/teacher/admin) with a repository-pattern data layer, a Python FastAPI microservice for scoring, and a queued job pipeline for asynchronous similarity checks
- Built a similarity report interface highlighting matched passages between submissions, with source attribution distinguishing peer matches from web matches

Secure Skills Hub - Learning Management System

2026

[Live](#)

- Designed a private, instructor-led LMS (Laravel, Alpine.js, Tailwind CSS) as a solo client project, replacing traditional marketplace-style LMS infrastructure with a focused, invite-only learning environment
- Built a token-based enrollment system to replace payment infrastructure entirely, giving instructors direct control over student access
- Enforced role-based access control at the middleware level across instructor, student, and admin flows, keeping route protection declarative rather than scattered through controllers
- Integrated Google OAuth for SSO, reducing signup friction and eliminating password storage; structured the codebase around the repository pattern to decouple business logic from the data layer

DwellCasa – Hotel Management System

2026

[Live](#)

- Designed a full-stack hotel management platform (Laravel, Alpine.js, Tailwind CSS) using the repository pattern to decouple business logic from data access across a multi-location tenancy model

- Implemented a status-based availability engine handling the full reservation lifecycle — bookings, check-ins, check-outs, and automated no-show resolution — preventing double-booking across concurrent requests
- Built a dual-track inventory system: consumable supplies with usage logs; durable equipment tracked by serial number, room assignment, condition history, and write-off status
- Enforced role-based access control across property locations — super admins switch between sites; admins and staff are scoped to their assigned property only